Questions

Q1.

Ultraviolet (UV) waves from the Sun travel towards the Earth.

Ultraviolet waves can be grouped by wavelength.

The three groups of wavelengths are UVA, UVB and UVC.

Figure 7 shows, for each group,

- the wavelength range
- the effect of the Earth's atmosphere on each type of UV wave.

	UVA	UVB	UVC
wavelength range	400 nm to 315 nm	314 nm to 280 nm	279 nm to 100 nm
% energy absorbed by the Earth's atmosphere	5%	95%	100%

Figure 7

(i) Explain why UVC is potentially the most dangerous ultraviolet radiation but does not cause harm to people.

(2)

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(ii) The speed of electromagnetic radiation is 3.00×10^8 m/s .

Calculate the frequency of the shortest wavelength of UVB radiation.

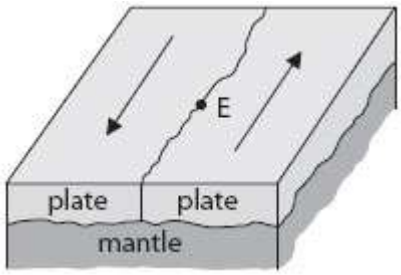
(3)

frequency = ... Hz

(Total for question = 5 marks)

Q2.

(a) The diagram shows part of the boundary between two tectonic plates.



(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

The plates are being steadily pushed in opposite directions by

(1)

☐ **A** convection currents in the mantle

☐ **B** reflection of waves from the Earth's core

☐ **C** tsunami waves in the ocean

☐ **D** volcanic eruptions on the surface

(ii) An earthquake occurs.
Its epicentre is at the place marked E on the diagram.
Describe what happens at the plate boundary to cause this earthquake.

(2)

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(b) The earthquake causes seismic waves.

(i) S waves are one type of seismic wave. They travel at 0.65 km/s.
There is a seismometer 80 km away from point E.
Show that it takes about 2 minutes for the S waves from the earthquake to reach the seismometer.

(2)

(ii) P waves are another type of seismic wave.
They travel about 10 times more quickly than S waves.

Describe how scientists can use seismometer records of P and S waves to locate the epicentre.

(3)

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(iii) Seismic waves have a frequency of about 15 Hz.
P waves have a much smaller amplitude than S waves.

Some people claim that animals can detect an earthquake before people are aware of it.
Suggest an explanation for this.

(2)

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(Total for Question is 10 marks)

Q3.

(a) Complete the sentence by putting a cross (☒) in the box next to your answer.
Waves from an earthquake are

(1)

☐ **A** transverse waves only

☐ **B** electromagnetic waves only

☐ **C** both transverse and electromagnetic waves

☐ **D** both transverse and longitudinal waves

(b) The Earth's surface is made up of many tectonic plates.
The interior of the Earth is a source of thermal energy.
Describe how this thermal energy can cause earthquakes.
You may draw a labelled diagram to help with your answer.

(3)

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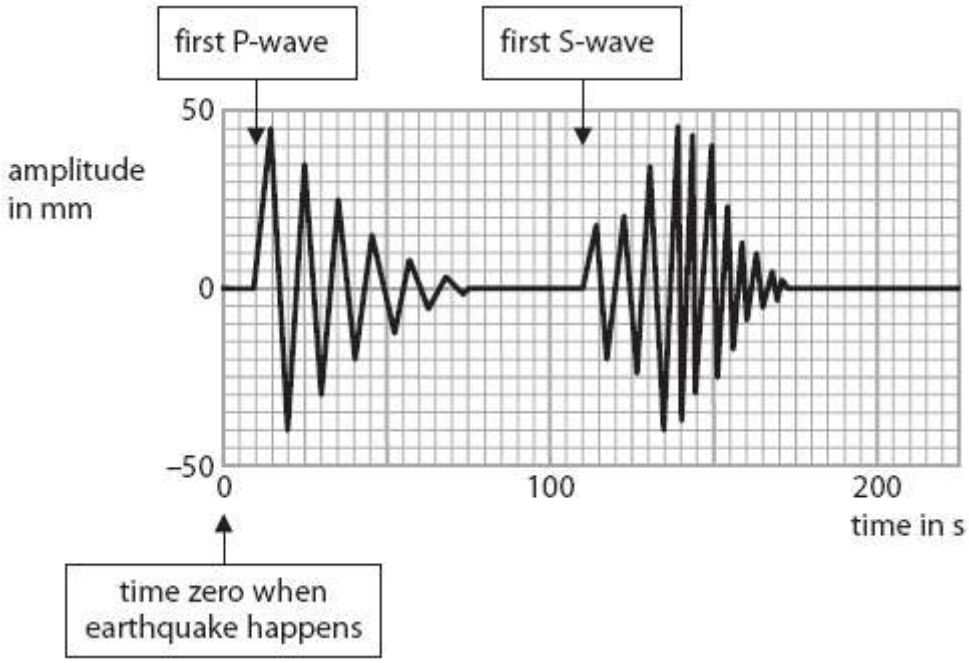
(c) The chart shows the arrival of earthquake waves at an earthquake monitoring station.



The S − P time (S minus P time) for earthquake waves is the time difference between the arrival of the first P wave and the first S wave.

Use the chart to estimate the S − P time for the earthquake waves shown.

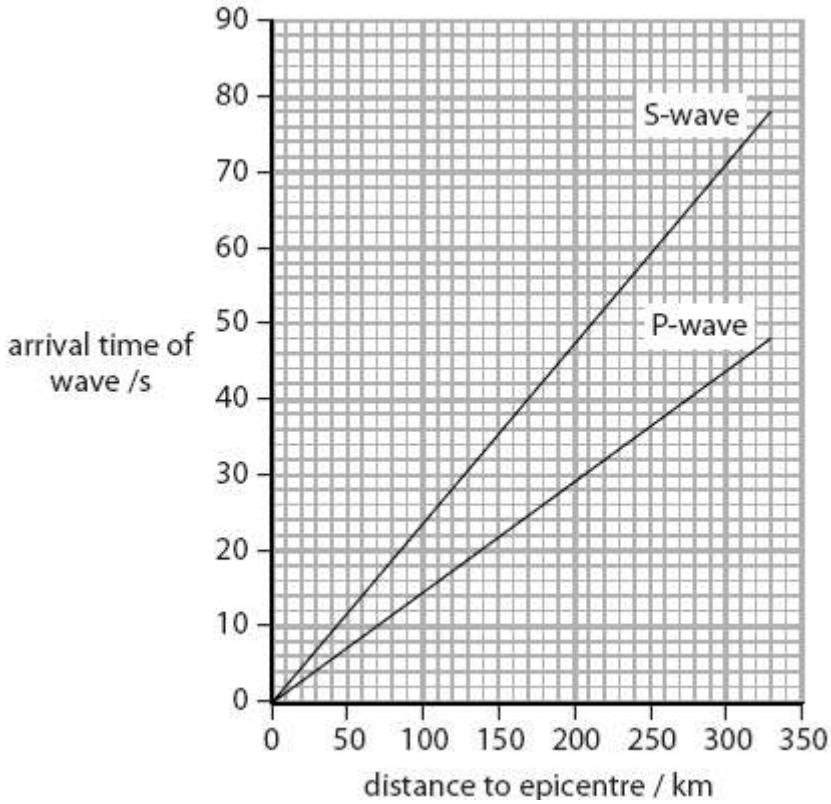
(2)

S − P time =.....seconds

(d) The location of an earthquake is known as an epicentre.

The S − P time for earthquake waves can be used to estimate the distance between the monitoring station and the epicentre.

The graph shows how the arrival times of S and P waves are related to their distances from the epicentre of an earthquake.



The S − P time for a particular earthquake was 20 seconds.
Use the S − P time to estimate the distance between the monitoring station and the epicentre of this earthquake.

(2)

distance to epicentre of earthquake = ...km

(e) Many earthquakes and volcanoes are linked to the production of infrasound waves.
Describe what is meant by **infrasound waves.**

(2)

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(Total for Question = 10 marks)

Q4.

Figure 8 represents a sound wave coming from a loudspeaker and shows the effects on particles of the air at one instant in time.

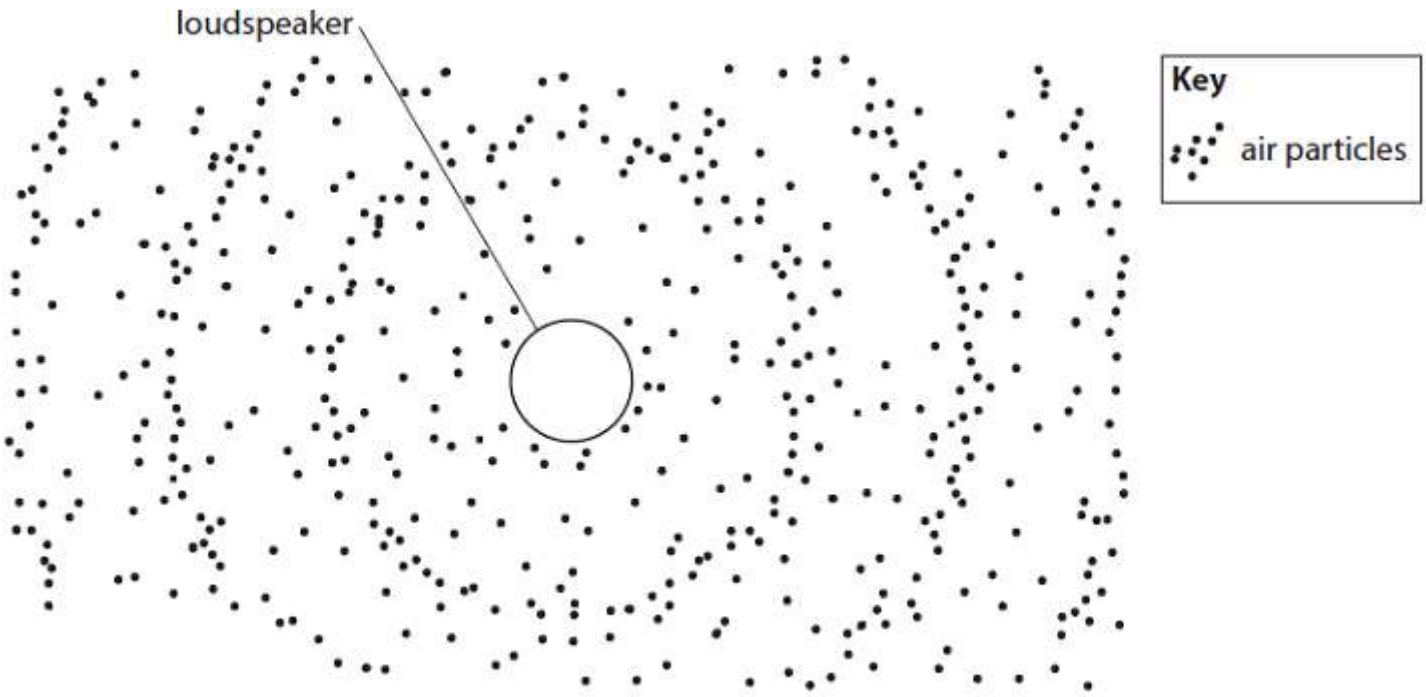

Figure 8

(i) Draw and label a distance of one wavelength in Figure 8.

(1)

(ii) Describe the motion of the particles as the wave travels through the air.

(2)

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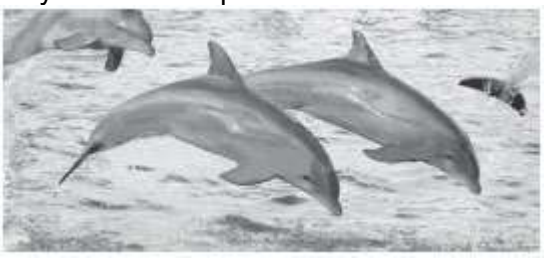
(Total for question = 3 marks)

Q5.

(a) Dolphins emit sounds at many different frequencies.



(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

A dolphin can produce sounds with a frequency of more than 20 kHz.
These sounds are called

(1)

☒ **A** infrasound waves

☒ **B** microwaves

☒ **C** supersonic waves

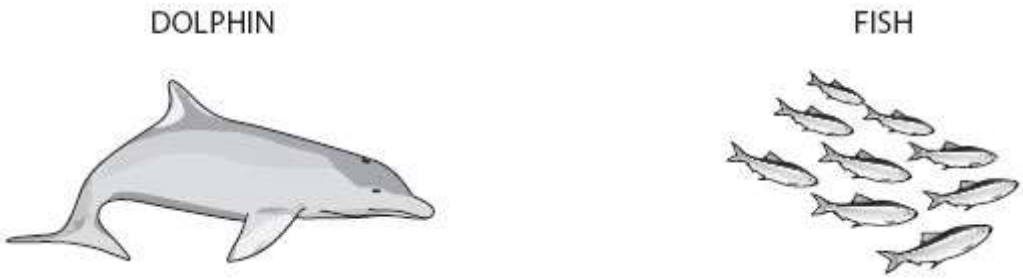
☒ **D** ultrasound waves

(ii) A dolphin is near to a group of fish.

Describe how a dolphin uses high frequency sound to locate the position of the fish.

You may add to the diagram to help with your answer.

(3)



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Q6.

Figure 5 is a diagram of part of a water wave.

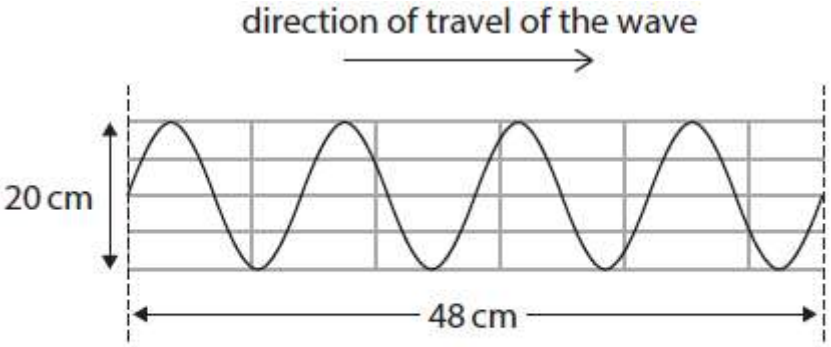

Figure 5

(i) Use data from Figure 5 to determine the amplitude of the wave.

(1)

amplitude = ... cm

(ii) Use data from Figure 5 to determine the wavelength of the wave.

(1)

wavelength = ... cm

(iii) The water wave shown in Figure 5 is generated in a large tank.

A student observes the wave.
Describe how the student could determine the wave speed of the wave.

(2)

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(Total for question = 4 marks)

Q7.

Describe the difference between 'infrasound' and 'ultrasound'.

(2)

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(Total for question = 2 marks)

Q8.

To investigate refraction in a rectangular glass block a student uses the apparatus shown in Figure 5.

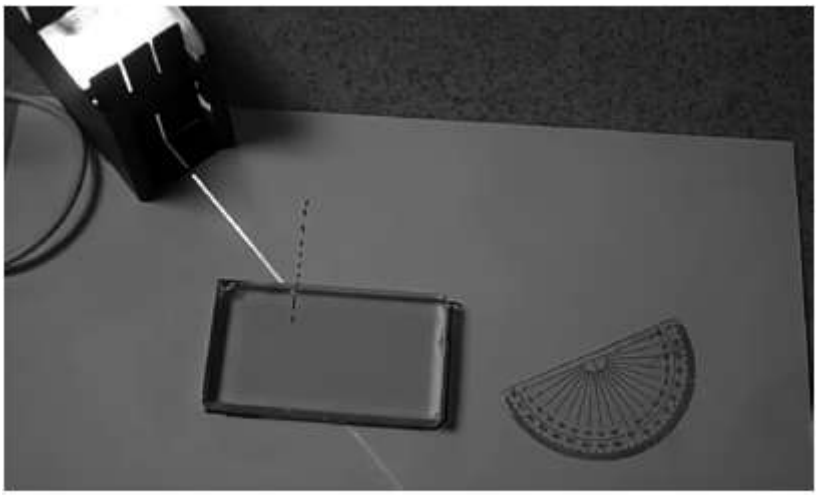

Figure 5

Describe how the student should measure the angle of refraction.

(2)

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(Total for question = 2 marks)

Q9.

Figure 8 shows a long metal rod and a hammer.
The rod is hit at one end by the hammer.
This causes a sound wave to travel along the inside of the metal rod.

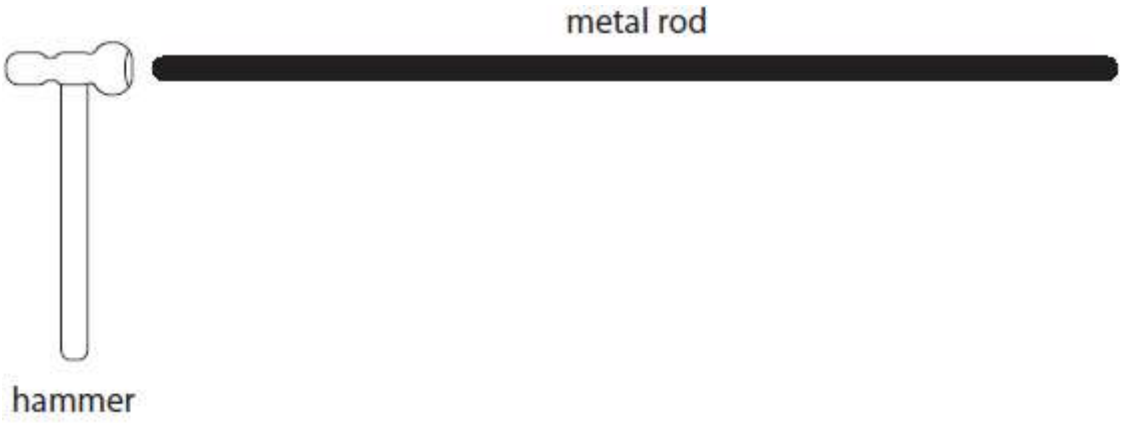

Figure 8

Describe how hitting the rod causes a sound wave to travel along the inside of the rod.

(2)

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(Total for question = 2 marks)

Q10.

Scientists believe that dolphins use sound waves to detect objects under water.

Describe how a dolphin could use sound waves to determine which of two objects is nearer to it.

You may draw a diagram to help your answer.

(3)

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(Total for question = 3 marks)

Mark Scheme

Q1.

Question	Answer	Additional guidance	Mark
(i)	An explanation linking UVC/it has the smallest wavelength / highest frequency/ highest energy/most ionising (1) (it doesn't cause harm to people because) 100% absorbed by the Earth's atmosphere/no UVC reaches the Earth's surface/people (1)	 allow shortest wavelength allow smaller for smallest / higher for highest / more for most	2 AO3.2

Question	Answer	Additional guidance	Mark
(ii)	wavelength 280(nm) (1) substitution and rearrangement (1) $f = \frac{3.0 \times 10^{8}}{280 (\times 10^{-9})}$ evaluation (1) 1.07×10^{15} (Hz)	allow 2.8 to any power of ten accept numbers that round up to 1.1 x 10^{15} (Hz) award 2 marks for values that round up to 1.1 to any other power of ten if no other mark scored allow substitution and rearrangement using an incorrect wavelength for 1 mark award full marks for correct answer without working	**3** **AO2.1**

Q2.

	Answer	Acceptable answers	Mark
(ai)	A		**(1)**
(aii)	A description linking plates move / slip / separate (relative to each other) (1) sudden (release of energy) (1)	plate rubs against each other friction between plates plate boundary shifts jerk / jolt	**(2)**

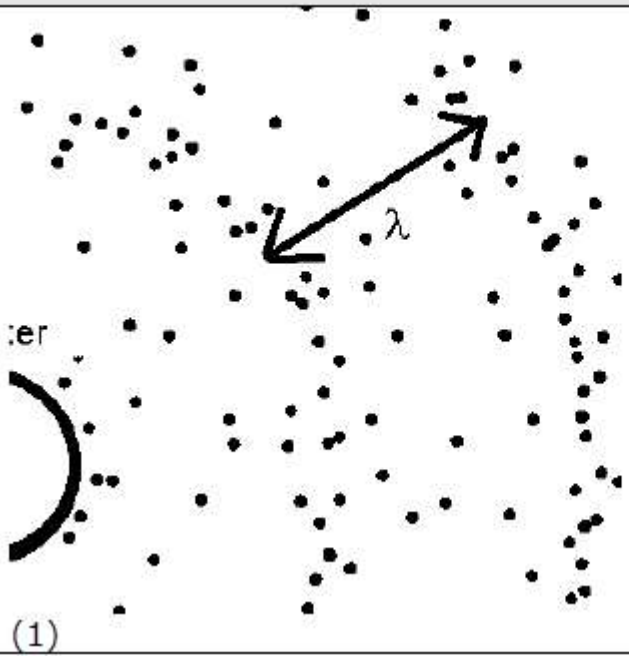
(bi)	substitution (1) 0.65 = 80 / t transposition (1) t = 80 / 0.65 (123 seconds)	transposition and substitution can be in either order . Allow reverse calculations eg speed = 80/120 (1) = 0.67 (about 0.65) (1) or distance = 0.65 × 120 (1) = 78 km (about 80) (1).	**(2)**
(bii)	A description linking any three detection of arrival of P and S waves (1) measurement of difference in arrival times (1) calculation of distance (from epicentre to station) (1) triangulation/using three / several stations (1)	Reward suitable labelled diagram	**(3)**
(b)(iii)	A suggestion including any two of the following Infrasound (1) some animals can hear waves below human frequency range / 20 Hz (1) they could hear P waves arriving before the (stronger) S waves arrive (1)	Some animals have greater audio / tactile sensitivity than humans	**(2)**

Q3.

	Answer	**Acceptable answers**	**Mark**
(a)	☒ **D** both transverse and longitudinal waves		**(1)**
(b)	A description including **three** of the following points • molten rock/magma (in mantle) (1) • convection currents (in mantle) (1)	Marks can be awarded on a labelled diagram Description of convection currents or arrows on diagram plates rub together Jolt/jerk	**(3)**

	• plates move (1) • build up of pressure/force/energy (when plates (not) sliding over/under/past (each other)) (1) • sudden movement when pressure becomes too great/is released (1) • This sudden movement of plates is an earthquake (1)	when pressure becomes too great/ is released	
(c)	relevant values 110 and 10 seen anywhere(1) 100 (s) (1) **acceptable range 95 to 105** (s)	(could be on chart) tolerance +/- 5 s give full marks for correct answer, no working	**(2)**
(d)	any vertical line of 10 squares on graph between P- wave and Swave (1) OR times eg 52 (s) – 32(s) Range (48 to 56) – (29 to 35) 220 (km) (1)	Range 9 to 11 squares range 200 to 240 (km) give full marks for correct answer, no working	**(2)**
(e)	a description including **two** of the following: • longitudinal / sound (wave) (1) • (frequency) less than 20 Hz (1)	Frequency below range/too low for (normal) human ear	**(2)**

Q4.

Question number	Answer	Additional guidance	Mark
(i)		any similar distance labelled wavelength / λ between the equivalent of 2 consecutive compressions	(1)

Question number	Answer	Additional guidance	Mark
(ii)	description including any two from: particles vibrate / oscillate/ move backwards and forwards (1) along a radius/ parallel to direction of travel/ energy transfer (1) about mean /fixed positions (1)	 allow air for particles in same direction as wave allow one mark for 'sound is a longitudinal wave' if no other mark awarded	(2)

Q5.

	Answer	Acceptable answers	Mark
(i)	D ultrasound waves (1)		**(1)**
(ii)	Any 3 from • emits (high frequency/ultra) (sound)(1) • (sound is)	Makes/sends out/produces (ultra sound/signal/wave) Uses 'high frequency sound' is insufficient	**(3)**

	reflected (off fish) (1) • (reflection) detected by Dolphin (1) • dolphin (estimates) time between (sending and receiving) sounds (1) • dolphin is able to change time into (estimate of) distance (1)	(sound) bounces off (fish) or echoes towards dolphin 1st three marks can be scored on the diagram. ie unless stated otherwise, assume any waves/rays starting at dolphin are ultrasound. Rays do not need to be straight	

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	10(.0) (cm)		(1) AO1.1

Question Number	Answer	Additional guidance	Mark
(ii)	12(.0) (cm)		(1) AO1.1

Question Number	Answer	Additional guidance	Mark
(iii)	a description to include		(2) AO1.2
	time a wave (crest) over a measured/fixed/known/stated distance (1)	measure how long it takes for a wave to travel a set distance measure distance travelled in a stated time	
	use (speed =) $\frac{\text{distance}}{\text{time}}$ (1)		
	ALTERNATIVE METHOD		
	(determine frequency by measuring) the time taken for a set number of waves to pass a point (1)	(count) the number of waves in a specified time 'measure frequency' is not sufficient for this mark	
	use (speed =) $f \times \lambda$ (1)	use (speed =) f x (0.)12	

Q7.

Question number	Answer	Additional guidance	Mark
	description including: infrasound < 20 Hz (1) ultrasound > 20 000 Hz (1)	award 1 mark for infrasound lower frequency than ultrasound OR reverse argument	**(2) grad**

Q8.

Question number	Answer	Additional guidance	Mark
	A description including two from: trace / mark where the ray went into and out of(through) the glass block / line on either side of glass block(1) (remove block) join entry and exit points (of ray of light) (1) use the protractor to measure the angle between the refracted/drawn ray and the normal (1)	accept 90° line etc.	**(2)**

Q9.

Question Number	Answer	Additional guidance	Mark
	A description including particles (at end) vibrate (more) (about fixed positions) (1) cause neighbouring particles to vibrate (more) (1)	allow atoms / ions / molecules for particles vibrations passed along OR reference to longitudinal waves / compressions and rarefactions	**(2)**

Q10.

Question Number	Answer	Additional guidance	Mark
	An answer that combines the following points of understanding to provide a logical description: • (dolphin emits and) receives sound reflected from the objects (1) • (dolphin determines) difference in times taken to receive the echo (1) • sound received in a shorter time from nearer object (1)	may be inferred from a diagram	**(3)**